

Modeling Global Temperature Variate Effects Over Time

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Big Picture

- Attempting to understand the relationship between climate variables (temperature, precipitation) and geography/time
 - Want to create a model for Temperature and Precipitation as a function of Latitude, Elevation, and Time

Motivation

- Understanding evolution of model parameters in modern climate
 - Given a model of climate, how do the coefficients change over time?
- Generate general model to apply to ancient data
 - Paleogene era: 66-23 Mya
 - Sparse proxy data (fossils, deposition layers, etc.)
 - $y(s, t) = x(s, t)^T \beta_t + z(s, t) + \epsilon(s, t)$
 - Can learn appropriate structure for design matrix X using modern data
- Put me through my paces

Method

ERA5 reanalysis dataset

- ERA5: 5th generation of ECMWF (European Centre for Medium-Range Weather Forecasts) reanalysis data for last 80 years [1]
- Reanalysis: Uses available data, models, and physical laws to provide a complete and consistent dataset of climate measurements [1]

Data Processing:

- Raw ERA5 data: .25x.25 Latitude grid, daily averages
- Aggregated to 1x1 Latitude, weekly averages
- Elevation: ERA5 2024 Geopotential, converted to elevation (m)

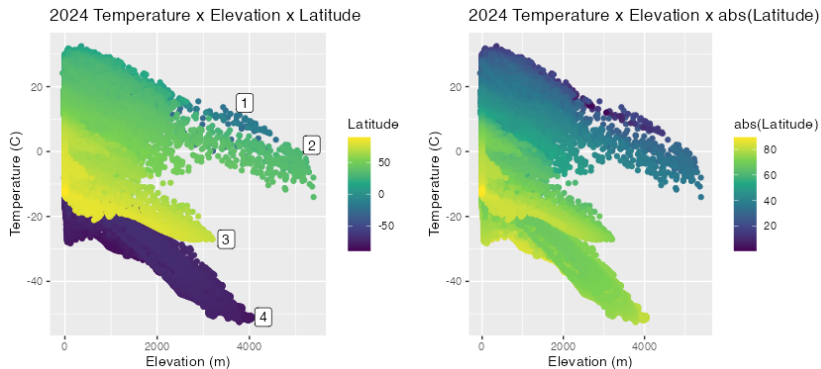


Figure: Numbered Regions: 1. Sierras 2. Himalayas 3. Greenland 4. Antarctica. Note the relatively linear effect of elevation, with some irregularities in the poles

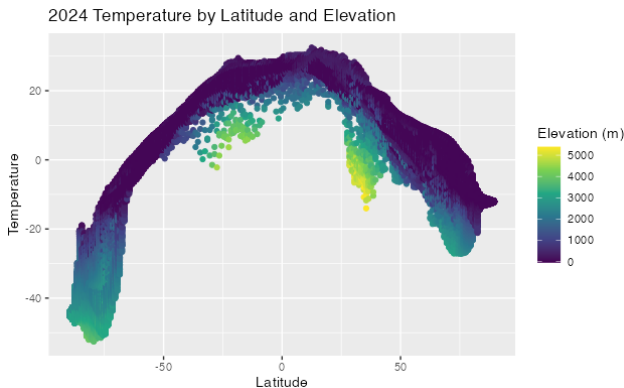


Figure: Temperature as a function of Latitude and Elevation interaction. Note the irregular 'rainbow' shape.

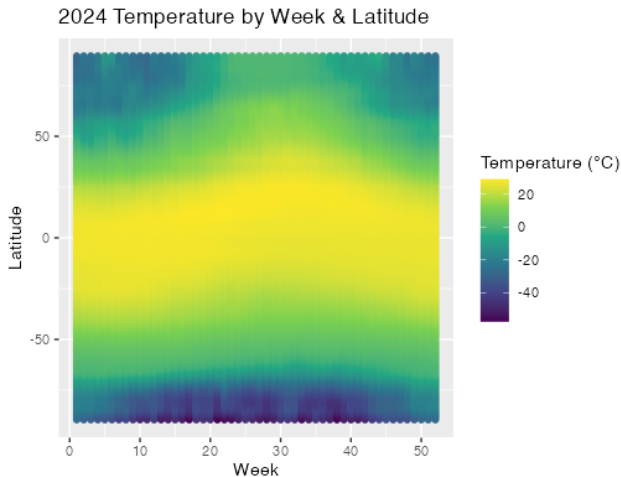


Figure: Temperature as a function of Week and Latitude. Has a 'macaroni' shape to the middle section, indicating the opposite behavior in the hemispheres. Has a sharp shift in temperature weeks 17, 45 in the Arctic

Model Requirements

Our model will need to account for the following:

- Interactions between each factor (Week, Latitude, Elevation)
- Non-uniform curve by Latitude
- Non-uniform curve by Week
 - Particularly opposite North/South Hemisphere behavior
- 'Week' is periodic (need $\text{Week}_{52_{y-1}}$ to map into Week_{1_y} smoothly)
- Non-centralized peaks in Latitude, Week

Model

- $y(s, t) = x(s, t)^{\top} \beta + \epsilon_{s,t}, \epsilon_{s,t} \overset{iid}{\sim} N(0, \sigma^2 I)$
- Design Matrix given by: $(bs(Lat) + ns(Elev) + bs(Week))^3$
 - Includes Latitude, Elevation, Week, all 2 and 3 way interactions
 - All terms are fit with splines
 - Boundary conditions enforced on Week: Week 0 = Week 52

Model Comparisons

- We are considering different models, not a single way to do that
- Model Validation/Model Comparison strategies
 - Summary metrics: AIC/BIC, etc. Adding lots of stuff tends to just make it better
 - Local Features: look at specific behavior of each model, residuals, etc.
 - Want a model that balances the two (ie. really good AIC is not good if it completely ignores a certain region of the earth)
 - Paleoclimate data: What level of complexity can that dataset handle?
Bias towards simpler model structures

Variable Fits

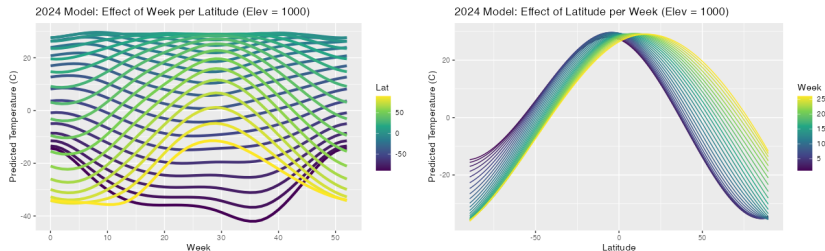


Figure: Predicted temperature as a function of Week and Latitude. Note (left) that min/max temperature is offset from middle of year. Also note (right) the steep decline in temperature in high latitudes.

Residual Analysis

- Evaluated all residuals across all weeks, found proportion of times the residuals at a location were above a set limit

Standardized Residuals Map - Stand. Res. Lim.:2.5

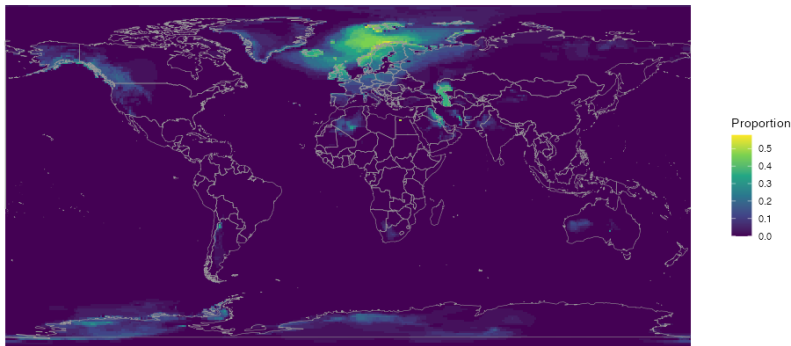


Figure: Proportion per location of residuals over a certain limit. Of interest are the Caspian sea, Norwegian Sea, coastal areas, mountain regions, and some deserts

Next Steps

- Additional Data
 - Coast Lines
 - Percentage of cell that is water
 - Deserts (proxy: NDVI satellite measure of vegetation)
- Model Improvements
 - Explore models, pick a final candidate
 - Account for spacial and temporal correlation
 - Massive dataset, look into Vecchia
- Further Analyses
 - Fit model to previous years, see how coefficients change
 - Apply similar techniques to Precipitation

Conclusion

- Obtained and aggregated ERA5
- Created model of temperature using Latitude, Elevation, Week
- Investigated residuals of model to identify features that still need to be addressed
- Future work will focus on collecting new data features, making improvements to model, and applying model to new years/climate variables

Bibliography

- [1] Copernicus Climate Change Service (C3S), *ERA5: Fifth generation of ECMWF atmospheric reanalyses of the global climate*, Climate Data Store (CDS), 2017. [Online]. Available: <https://cds.climate.copernicus.eu/cdsapp%5C#!/home>.